INCH-POUND

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SUPERSEDING
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DETAIL SPECIFICATION

COATING COMPOUND, BITUMINOUS, SOLVENT, COAL-TAR BASE

Inactive for new design after 3 February 2009

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a cold-applied coal-tar base coating compound for use on steel structures under conditions where it is not feasible to use a hot-applied coal-tar enamel.

2. APPLICABLE DOCUMENTS.

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARD

FED-STD-141	Paint, Varnish, Lacquer and Related Materials: Method of Inspection, Sampling and Testing
FED-STD-313	Material Safety Data, Transportation Data, and Disposal Data for Hazardous Materials Furnished to Government Activities

(Copies of these documents are available online at https://assist.dla.mil.)

Comments, suggestions, or questions on this document should be addressed to: DLA Troop Support-Construction and Equipment (C&E) (ATTN: Code FCSB), 700 Robbins Ave, Philadelphia, PA 19111-5096 or email to CE_Standardization@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 8030

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Laws and Regulations.

29 CFR 1910	Occupational Safety and Health Administration (OSHA), Department of Labor

(Copies of these documents are available online at https://www.osha.gov)

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS STANDARD (ASTM)

ASTM D5/D5M	Standard Test Method for Penetration of Bituminous Materials
ASTM D36/D36M	Standard Test Method for Softening Point of Bitumen Ring-and-Ball Apparatus)
ASTM D92	Standard Test Method for Flash and Fire Points by Cleveland Open Cup Tester
ASTM D128	Standard Test Method for Analysis of Lubricating Grease

(Copies of these documents are available from https://www.astm.org/ or the American Society for Testing and Materials, 100 Barr Harbor Drive, W. Conshohocken, PA 19428-2959.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The compound shall be a homogeneous mixture composed only of a coke-oven coal-tar pitch, solvents, and inert non-water-absorbent mineral filler. The compound shall not contain asphalt or asbestos.

3.1.1 Solvent composition. When specified, the compound shall conform to applicable air quality control regulations as specified (see 6.2).

3.2 Requirements. The compound shall to the requirements given in Table I.

3.3 Toxicity. The compound shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the acquiring activity to the appropriate medical service who will act as adviser to the acquiring activity. The manufacturer's instructions shall provide personnel protection to meet OSHA requirements, including 29 CFR 1910.1000, and 1910.1002, as applicable. Material Safety Data Sheets (MSDS) shall be prepared in accordance with FED-STD-313, and submitted, as directed in the contract or order, at the time of acquisition award. Copies shall be forwarded to the designated Industrial Hygienist and the focal point of the activity that purchased the item, and the focal point of the using activity if different (see 6.2 and 6.3).

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TABLE I. Requirements.

Property	Characteristics Min	Max	Test Method	Test paragraph
Flash Point, °C	35	---	ASTM D92	4.1.1.1
Penetration of distillation residue, mm	5	25	ASTM D5/D5M, 235°C residue sample	4.1.1.1
Softening point of distillation residue, °C	96	115.5	ASTM D36/D36M, 235°C Residue sample	4.1.1.1
Ash content of Compound, wt. %	15	30	ASTM D128, rapid routine method	4.1.1.1
Workability	Satisfactory working and spreading		---	4.1.1.2
Sag	No sag or flow while wet		---	4.1.1.3
Drying 6 hours 24 hours	 Set-to-touch Dry-to-recoat		FED-STD-141, Method 4061.3 application at 120 ft^2/gallon	4.1.1.1
Adhesion and protection	Shall not loosen, check, crack, peel, run, sag, or otherwise lose protection value		---	4.1.1.4
Resistance to impact	No visible chipping, cracking, or detachment from plate, and firm adhesion outside radius of 0.25 inch from center of impact		---	4.1.1.5
Resistance to alkali	No evidence of disintegration		---	4.1.1.6

4. VERIFICATION

4.1 Conformance inspection. Quality assurance shall be provided in accordance with method 1031.1 of FED-STD-141. Failure of the coating compound to pass any test shall be cause for rejection of the lot.

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4.1.1 Tests.

4.1.1.1 Requirements. The coating compound shall be tested by the methods specified in Table I.

4.1.1.2 Workability. Test the coating compound, applied cold, for firm adherence to bare steel, primer (if used), and itself. At 25 ± 2° Celsius (C), the coating shall permit easy application by brush or spray, in two successive coats, to a minimum dry film thickness of 30 ± 2 mils. At 7 ± 2ºC, the coating shall permit easy brush application.

4.1.1.3 Sag. At 23 ± 1°C and 50 ± 4 percent relative humidity, apply the compound to a clean, smooth 12 by 12 by 0.125-inch thick steel plate, to a uniform wet film thickness of 30 ± 2 mils. Immediately after application, suspend the panel in a vertical position at application conditions for 24 hours. Then examine for evidence of sag or flow while wet. Prepare a second plate as described, except to a uniform wet film thickness of 15 ± 1 mils. Immediately after application, suspend the panel in a vertical position at 71 ± 2°C for 1 hour. Then examine for evidence of sag or flow while wet.

4.1.1.4 Adhesion and protection. Expose the coated panel, tested at 23°C in 4.1.1.3 to 60 ± 2°C for 16 hours. Allow the panel to cool at room temperature 1 hour, then expose to -23 ± 2°C for 4 hours. Examine the coating for conformance to Table I.

4.1.1.5 Resistance to impact. Apply the compound to two wire-brushed, solvent-cleaned, mild steel plates, each 6 by 6 by 0.125-inch thick, to a uniform dry film thickness of 30 ±2 mils. Dry 72 hours at room temperature then test each plate separately, while being held firmly, coated side up, on a solid horizontal base. Drop a 2-pound steel ball from a height of 8 feet so that the impact will be at the center of the plate. Examine the coating for conformance to Table I.

4.1.1.6 Resistance to alkali. Apply the compound to a clean glass panel at a uniform dry film thickness of 30 ± 2 mils. Dry the panel at 23 ± 1°C for 1=24 hours, then suspend vertically in 5 percent sodium hydroxide, maintained at 23 ± 1°C, so that one-half of the coating is immersed. After 30 hours, lightly rub the film with a well-rounded glass rod and examine for evidence of disintegration.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contracting the responsible packaging activity.

6 NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The compound is intended for use on steel structures as necessary to substitute for hot-applied coal-tar enamel coating. The compound is used also as a protective coating for dissimilar metals in contact, as an electrical insulation coating, and on fiberglass lagging for waterproofing. It is appropriate for contact with alkaline soils, and may be used on underwater marine structures. Generally, two coats are used to a dried film thickness of 20 to 40 mils. Material to be exposed to sunlight or weather should be top-coated with coal-tar emulsion.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Applicable air quality control regulations, when specified (see 3.1.1).
c. Addresses for submission of MSDS (see 3.3 and 6.3).
d. Packaging requirements (see 5.1).

6.3 MSDS submission and forwarding. After review and acceptance of MSDS by designated recipients, approved copies will be forwarded to arrive at destinations prior to material delivery (see 3.3).

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.5 Subject Term (key Word) Listing:

Adhesion
Drying
Resistance to alkali
Resistance to impact
Sag
Workability

Custodians:
Army - MR
Navy - YD
Air Force - 99
DLA - CQ

Review activities:
Army - CE
Navy - SH

Preparing activity:
DLA - CQ

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